

LineProof

**Make robot progress
verifiable.**

A contact-verified SO-101 simulation
prototype with an Intel OpenVINO
runtime preflight.

CURRENT STATUS: SCRIPTED PROTOTYPE



A plausible ending can hide a failed grasp.

A cube can slip out, land near its target and still pass an endpoint-only check.

LineProof checks the journey:

- 01 Held by both jaws
- 02 Carried above the table
- 03 Lowered with support
- 04 Released and stable

Two arms. Real contacts. Auditable traces.

SIMULATION

MuJoCo + pinned SO-101
assets
Two cameras, 12 joint
targets
Contact-driven object
motion

CONTROL

Scripted IK teacher
Finite and bounded
commands
Stop/reset with stale-action
rejection

VERIFICATION

Versioned scoring
State/action evidence
Recorded failure
regression

Left arm manipulates one cube. Right arm is parked. IK uses simulator state.

MEASURED ON 16 SEPTEMBER

A verified baseline, with visible limits.

79.54 mm

minimum sustained lift in the nominal trial

9.14 mm

final horizontal placement error

6 / 10 passed

Small cube perturbations: XY ± 5 mm; yaw ± 0.12 rad. Four failures retained.

OpenVINO runs the ACT model on Intel CPU.

42.649 ms

OpenVINO median / five measured calls

Native PyTorch: 60.634 ms

FP32 agreement: max absolute error 0.000001386

Intel i5-1245U · four CPU threads · fixed input shapes

Random weights. No model action executed. This proves runtime compatibility only.

The learned full-task system is unfinished.

DELIVERED

Reproducible contact simulation
Nominal pick/carry/release recording
Ten perturbation reports + failure trace
13 passing tests + OpenVINO preflight

NOT YET DEMONSTRATED

Task-trained policy or VLM planning
Model-driven robot actions
Coordinated table-setting sequence
Physical robot or Core Ultra execution

Next gate: collect verified demonstrations → train a bounded policy → test closed-loop control.

Every claim has a trace.

github.com/IZAQ18/infra-summit

Start: README.md and docs/LOCAL_FEASIBILITY.md

Manipulation: docs/CONTACT_EXPERIMENT.md

Measurements: BENCHMARKS.md + docs/evidence/

Licenses: MIT code; Apache-2.0 upstream SO-101 assets

LineProof · Ahsan Maqbool / IZAQ

Reproducibility first. No completed-task claim.